Fiscal Year:	FY 2025	**Task Last Updated:**	FY 08/12/2025
PI Name:	Garbino, Alejandro M.D., Ph.D.		
Project Title:	Aerospace Estimation Tool for Hypobaric Exposure Risk Studies: Optimizing DCS Management for Lunar EVAs		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **EVA**:Risk of Mission Impacting Injury and Compromised Performance and Long-Term Health Effects due to EVA Operations		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 80JSC024NA001-FLAGSHIP2 2024 HERO Appendix E: NASA Human Research Program Flagship2 Opportunity
Start Date:	08/01/2025	**End Date:**	07/30/2031
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Estep, Patrick (NASA Johnson Space Center) Marshall-Goebel, Karina (NASA Johnson Space Center) Norcross, Jason (KBR/NASA Johnson Space Center) Thoolen, Stijn (KBR/NASA Johnson Space Center) Young, Millennia (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The transition from science-focused International Space Station (ISS) operations, where parameters replicate Earth environments, to NASA's upcoming exploration-focused Artemis missions, where the environment is focused on conducting extravehicular activities (EVAs), requires a significant departure from previous operational paradigms. To maximize EVA time and capabilities, lower-pressure environments are critical, but are accompanied by significant physiological challenges. The nitrogen in the environment is apt to cause venous gas emboli (VGE) "bubbling" and potentially decompression sickness (DCS) if the transition to a lower pressure (i.e., the spacesuit) is too abrupt. Denitrogenation protocols using a 100% oxygen prebreathe can mitigate this risk, but the specific prebreathe duration and methods vary depending on the model used. The primary models used to estimate DCS risk and calculate prebreathe duration are the Tissue Ratio (TR) model, based on the gas pressures in tissue compartments, and the Tissue Bubble Dynamics Model (TBDM), which uses tissue gas pressure to calculate the theoretical size of a bubble in tissue. Given the different natures of these models, the estimates of DCS risk, even when controlling for initial TR, can be widely different. Here, we aim to enhance our understanding of the underlying physiological mechanisms of DCS, to enable more accurate models for predicting its risk across various spaceflight mission parameters/vehicles; and to investigate means for enhancing denitrogenation (and thereby shortening prebreathe durations), allowing for adequate management of operational DCS risk and optimization of mission timelines. Specific Aims: Aim 1: Resolve the discrepancy between DCS prediction models (e.g., TR and TBDM models), by determining safe and operationally relevant tissue ratio thresholds for spaceflight and surface EVA DCS risk modeling and prediction. Aim 2: Evaluate the impact of increased perfusion, through targeted exercise levels and CO2 exposure on DCS and VGE incidence, to decrease necessary prebreathe duration during a simulated EVA. Aim 3: Determine the ability of intermittent recompression to reduce DCS and VGE incidence during a simulated EVA. Methods: A series of human subject hypobaric chamber tests will be run with incidence of DCS and quantification of venous gas emboli as the primary and secondary endpoints, respectively. All data will be input into a standardized database to enable DCS prediction modeling for the Virtual NASA Specialized Center of Research (VNSCOR) project.
Rationale for HRP Directed Research:	This project is directed because it contains highly constrained research.
Research Impact/Earth Benefits:	This approach will provide mission planners with a more accurate models of DCS risk, given certain mission parameters, as well as enhanced strategies for mitigating DCS in the most efficient ways possible for future exploration missions, while maximizing surface EVA time.
Task Progress:	New Project for FY2025
Bibliography Type:	Description: (Last Updated: 04/28/2026)